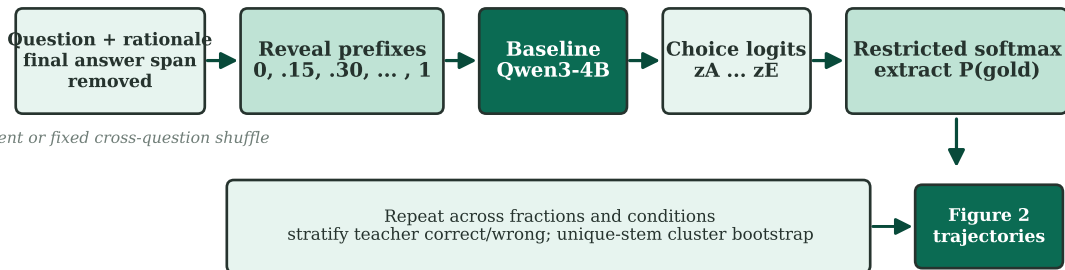


A Pre-training probability tracing



B Answer-verified training targets

Pure SFT

Answer X (1)

Proposed
distill-SFT

rationale (alpha)

Answer X (1)

Step-by-step

label task:
Answer X

rationale task
(alpha)

Prompt tokens are masked.
Only answer-verified rationales carry rationale loss.

C Loss-mass spacing

$$\text{answer share} = n_A / (\alpha n_R + n_A)$$

$\alpha = 1$ 2% answer

$\alpha = 0.5$ 4% answer
not evaluated

$\alpha = 0.3$ 6% answer

$\alpha = 0.1$ 17% answer

$\alpha = 0.3$: about $3 \times$ the answer share of $\alpha = 1$
 $\alpha = 0.5$: only about $2 \times$